BRAC UNIVERSITY

Inspiring Excellence

CSE422 Lab Project Report

Customer Category Classification

Semester: Spring 26

Section- 21

Group no-5

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1. Introduction

This project aims to predict customer segments based on customer characteristics and behavioral data. The target variable consists of four distinct categories: Segment A, B, C, and D. The motivation behind this project is that businesses can use such a model to better understand their customer base, optimize marketing strategies and deliver personalized services.

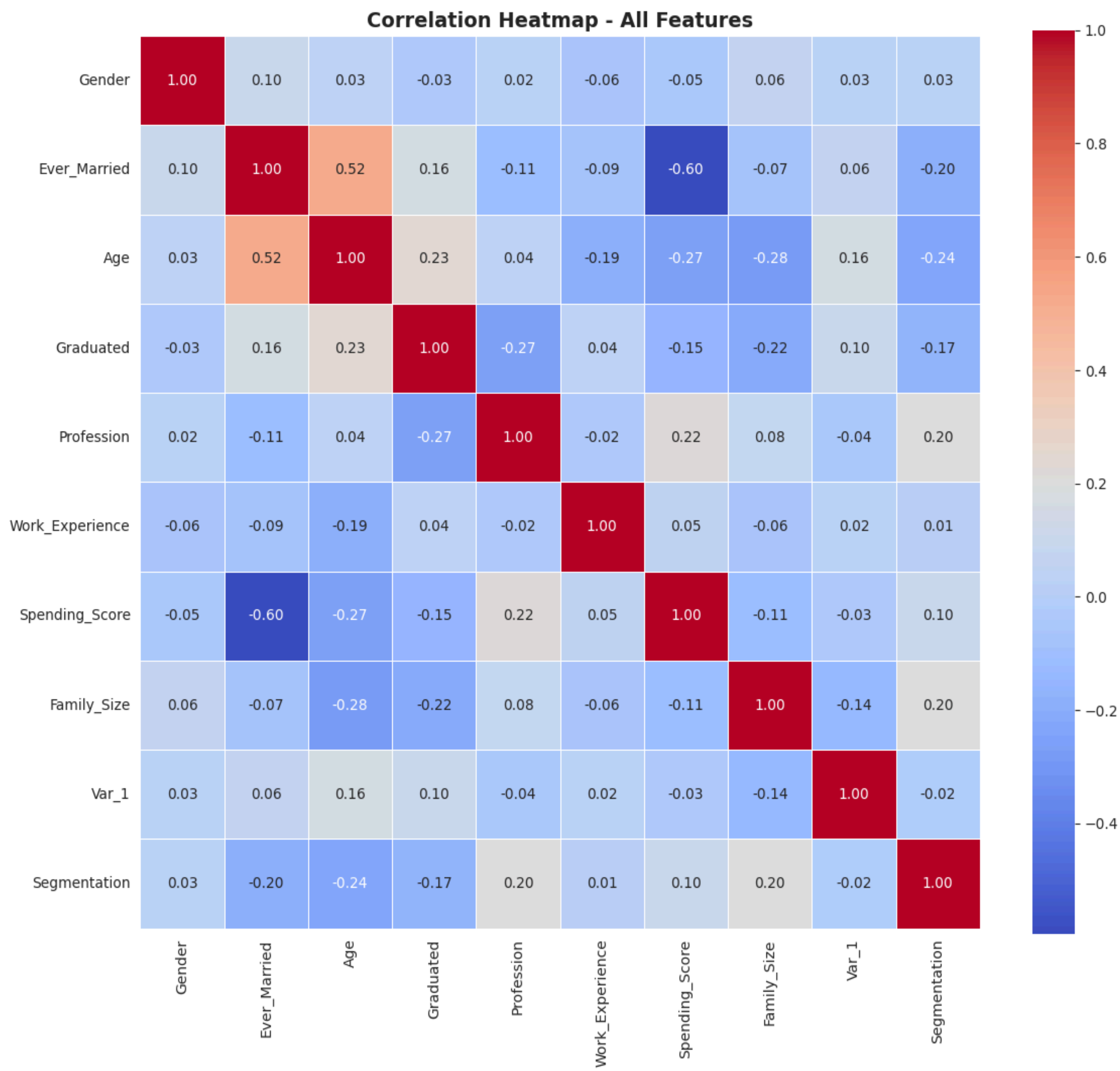
The objective of this project is to apply machine learning models to solve the problem of customer segmentation and explore the effectiveness of the models. This project follows a systematic approach from exploratory data analysis and data preprocessing to model training and evaluation, implementing multiple supervised and unsupervised learning techniques. This report presents a detailed description of the workflow.

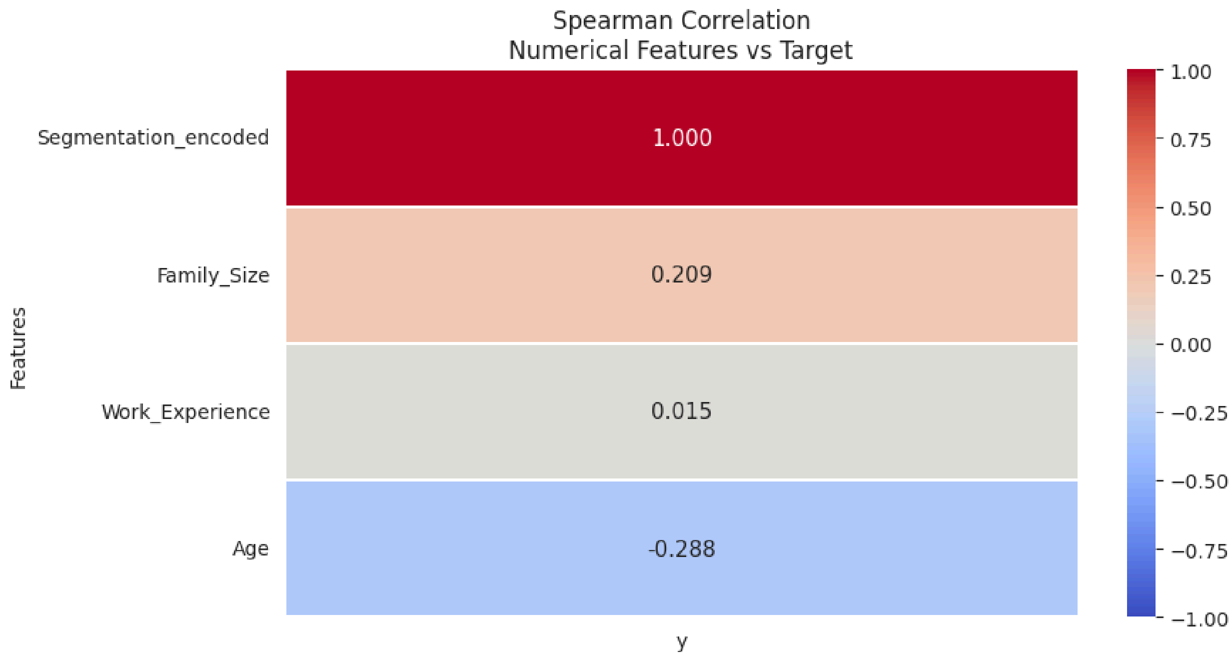
2. Dataset Description

2.1 Basic Information

- **Data Points and Features:** The dataset contains 8,068 data points and 11 features in total.
- **Problem Type:** This is a multi-class classification problem. The objective is to identify which of the four distinct segments (A, B, C, or D) a customer belongs to. Because the output variable is a discrete category rather than a continuous numerical value, it is a classification problem.
- **Feature Types:** The dataset contains both quantitative and categorical data:
    - **Quantitative or Numerical Features (4):** ID, Age, Work_Experience, Family_Size.
    - **Categorical Features (7):** Gender, Ever_Married, Graduated, Profession, Spending_Score, Var_1, and the target feature, Segmentation.
- **Categorical Encoding:** Yes, the categorical variables have to be encoded. Machine learning algorithms require numerical inputs to perform mathematical computations. The models can not understand string type data. Converting these textual labels into a numerical format is a necessary step so that the models can effectively process the data.

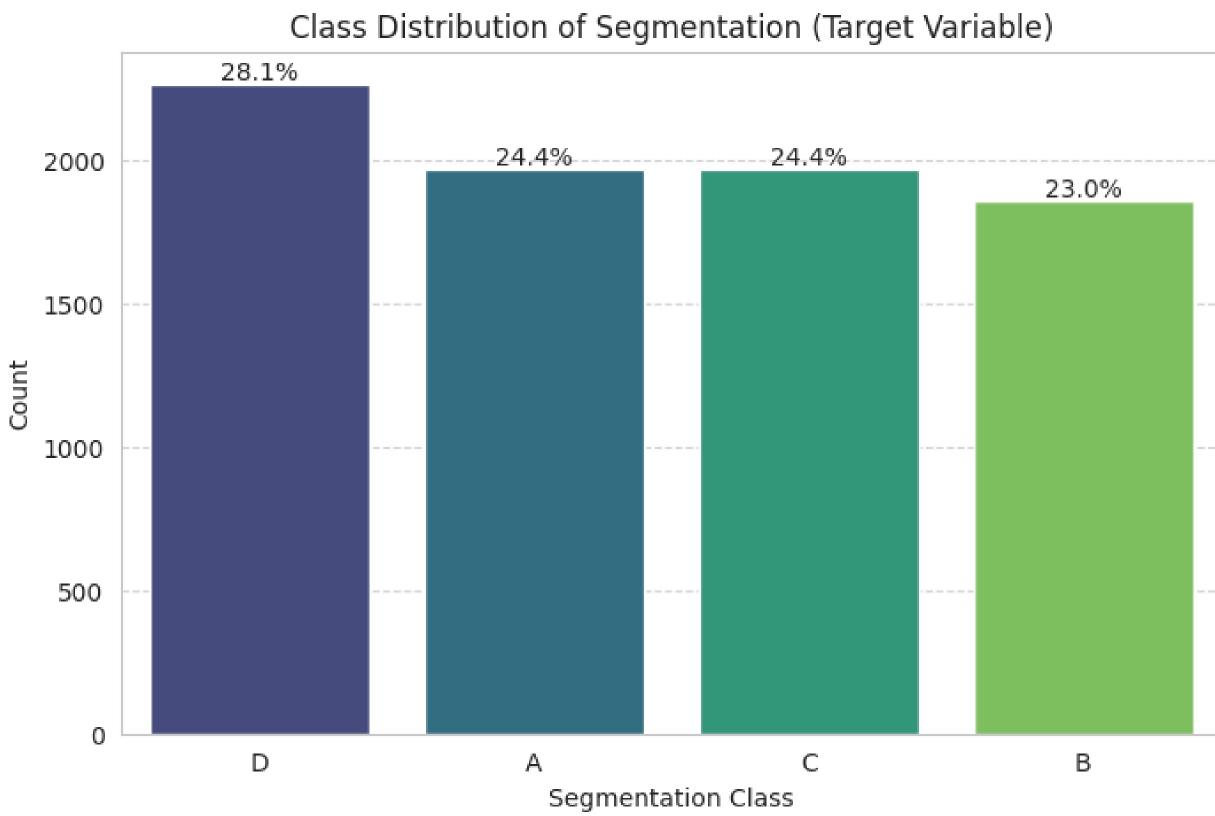
2.2 Correlation Heatmap of all the Features





Interpretation: There exists a very weak correlation between input variables and target variable segmentation as seen from both pearson and spearman correlation graphs. However, among the input features, age is the most and Work_Experience is least correlated with segmentation. Overall, the relationship between input and output variables is not linear and all features combined affects the segmentation.

2.2 Imbalanced Dataset Check

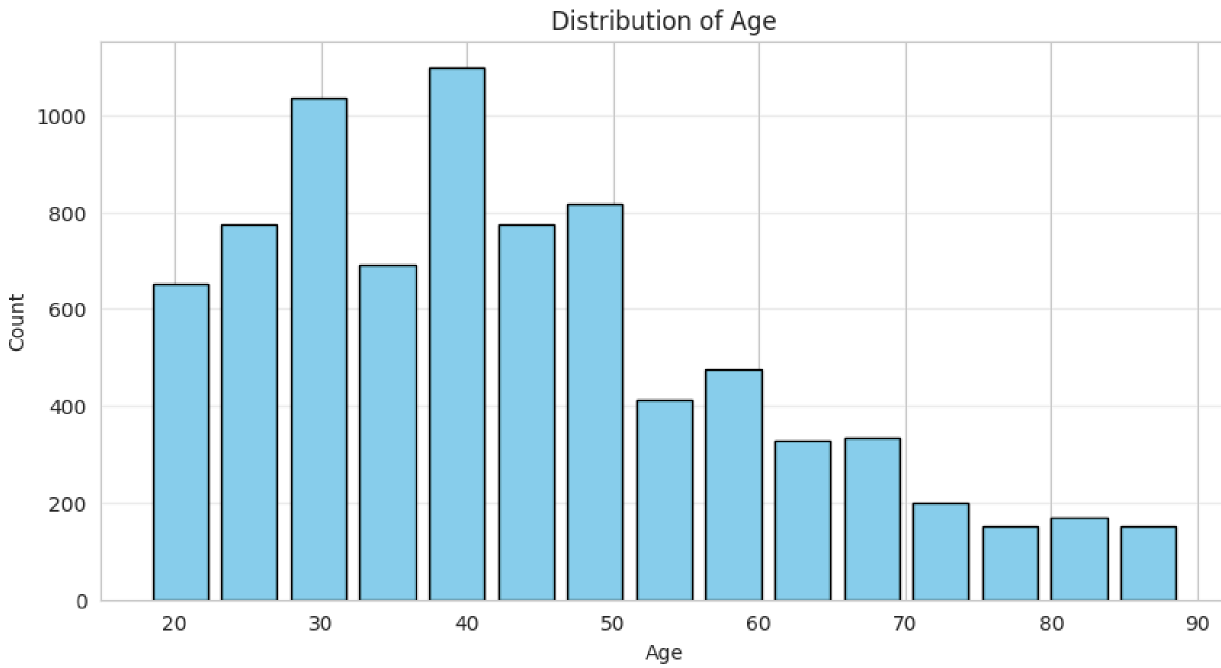


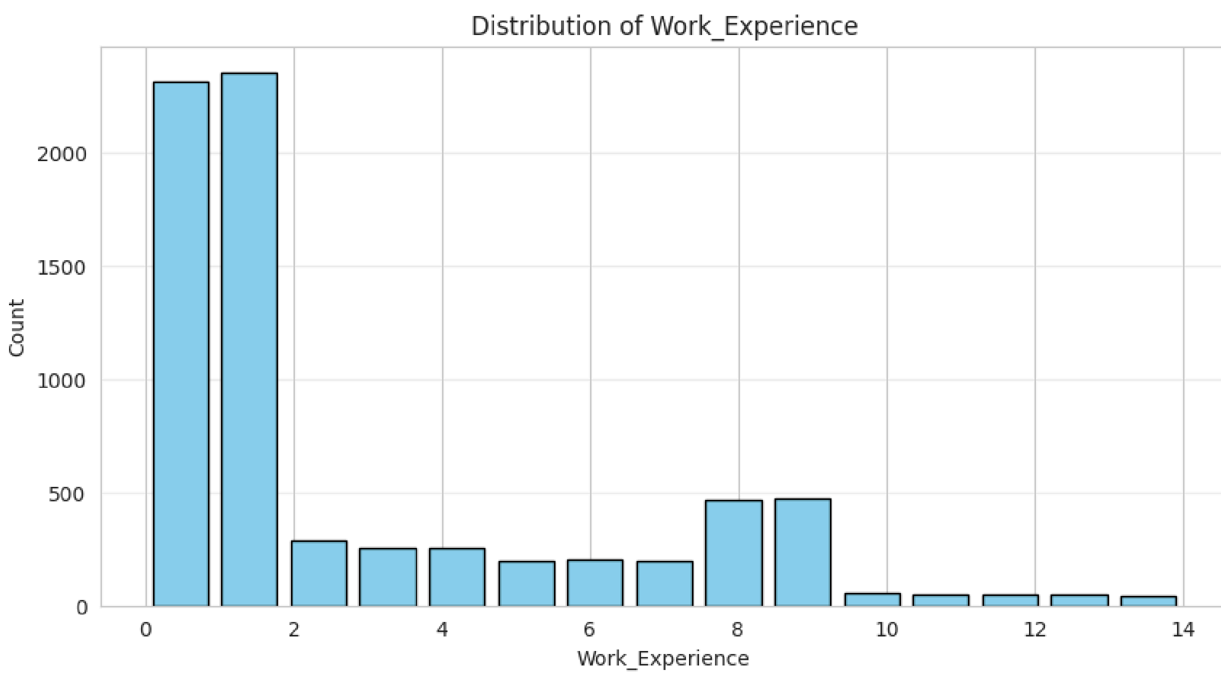
- **Segment D:** 2,268 instances (~28.11%)
- **Segment A:** 1,972 instances (~24.44%)
- **Segment C:** 1,970 instances (~24.42%)
- **Segment B:** 1,858 instances (~23.03%)

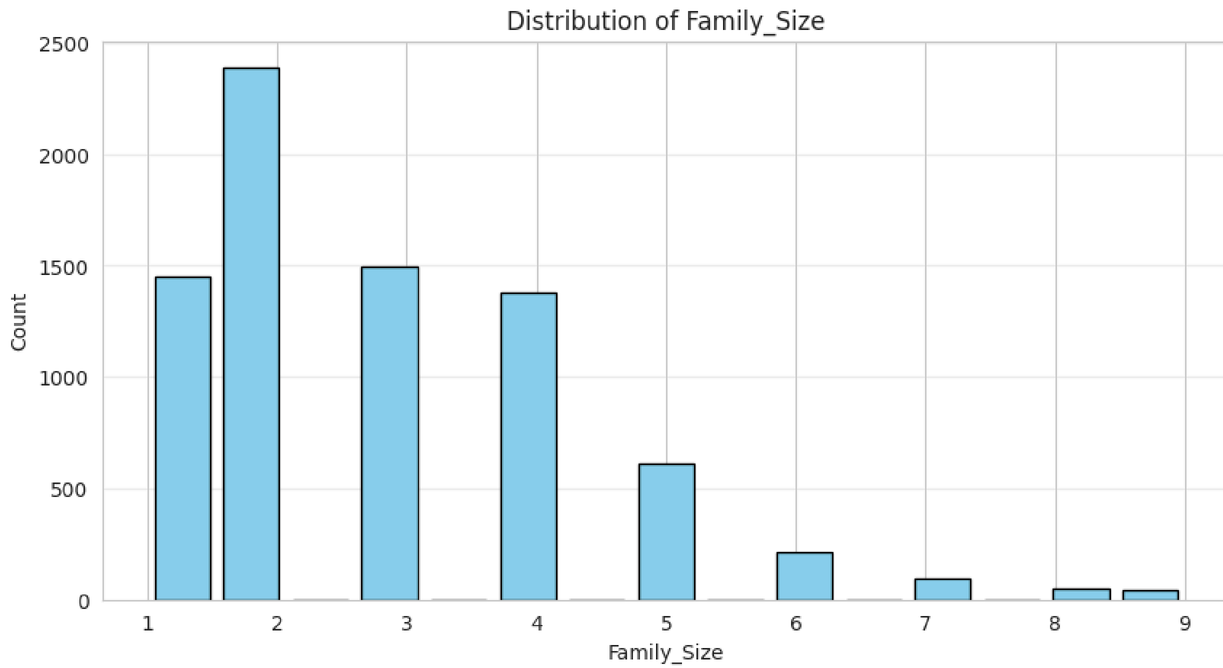
All unique classes do not have an equal number of instances as shown in the graph. But still the dataset is fairly balanced. Because each of the 4 classes have approximately 25% participation. So, models are less likely to have bias on a single class.

2.3 Exploratory Data Analysis (EDA)

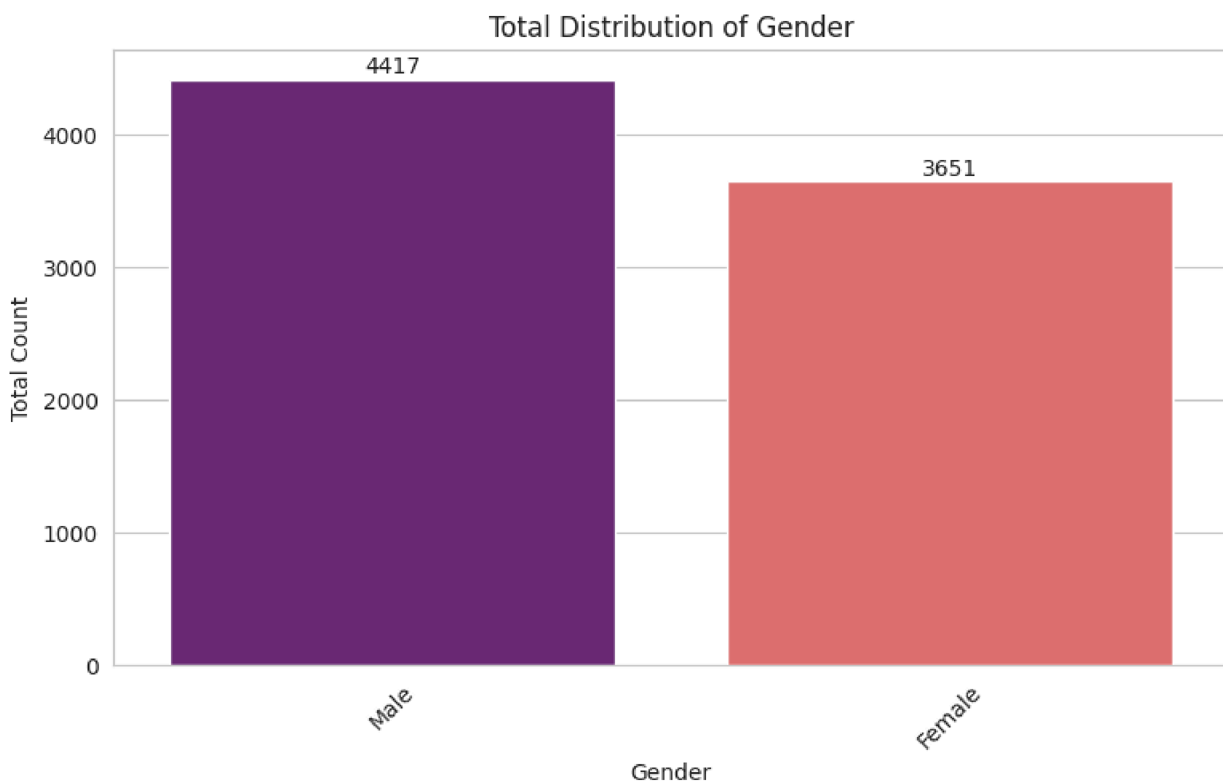
Barchart distribution of Numerical Features:

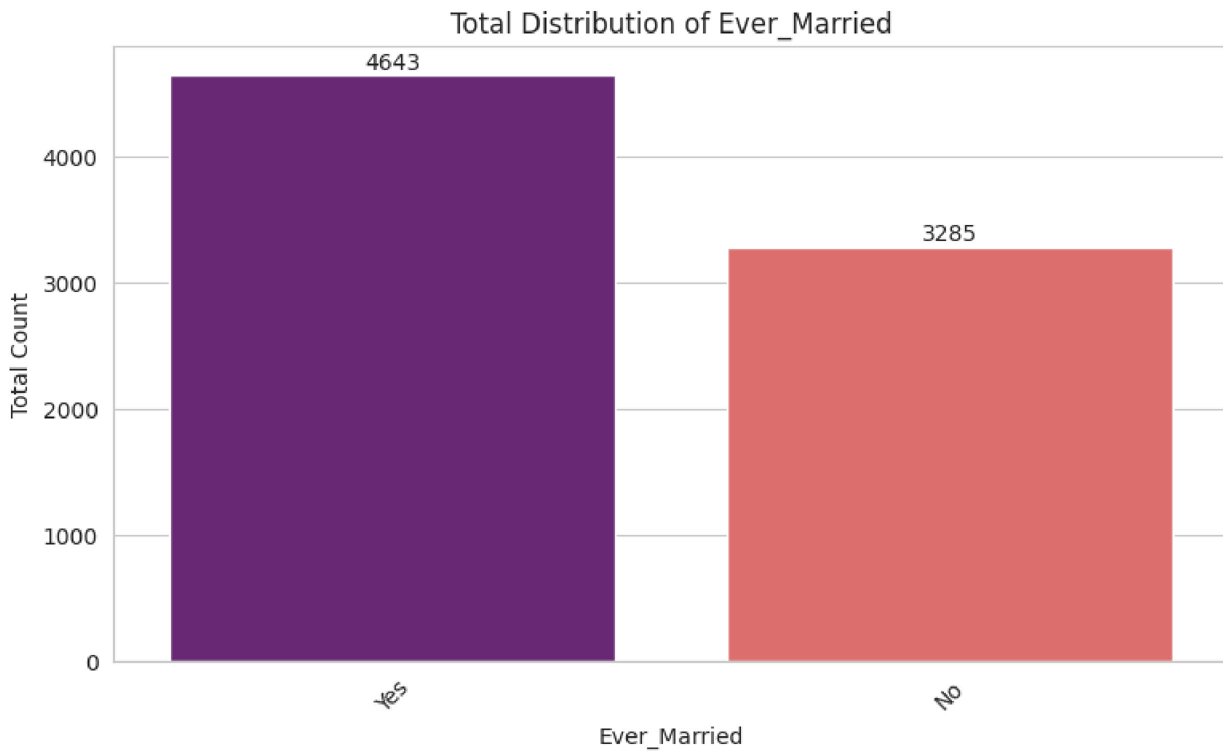


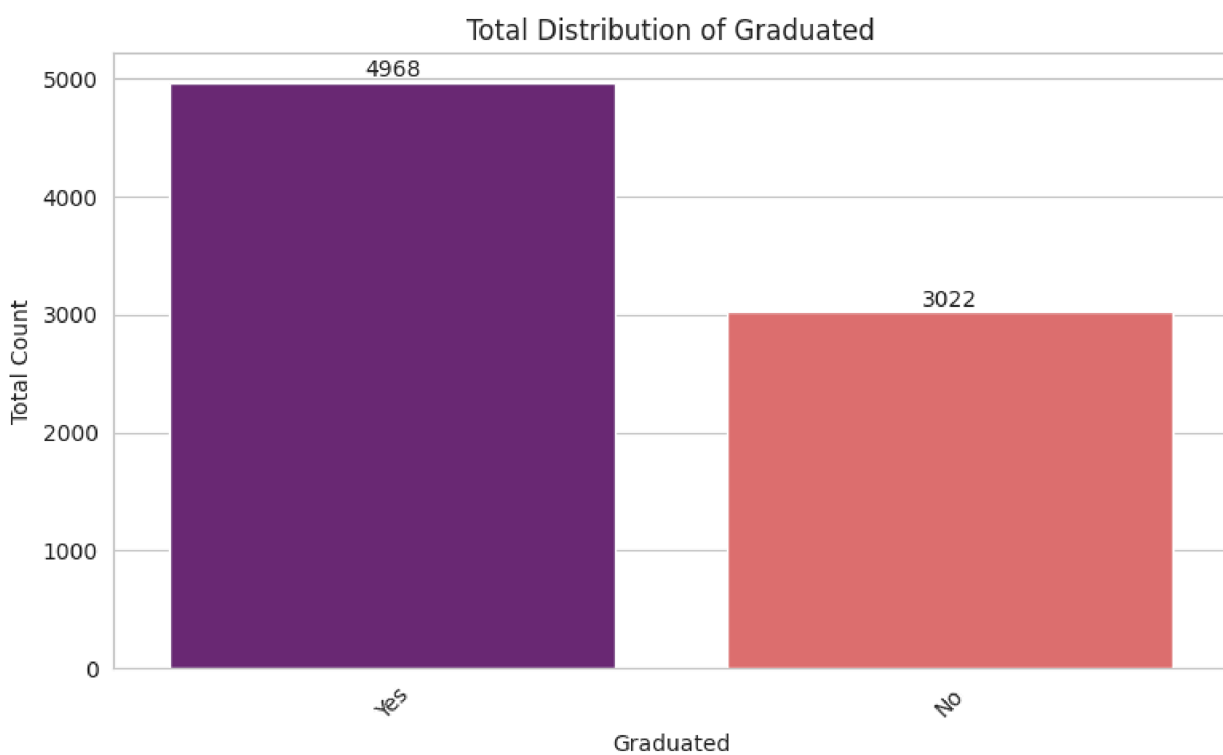


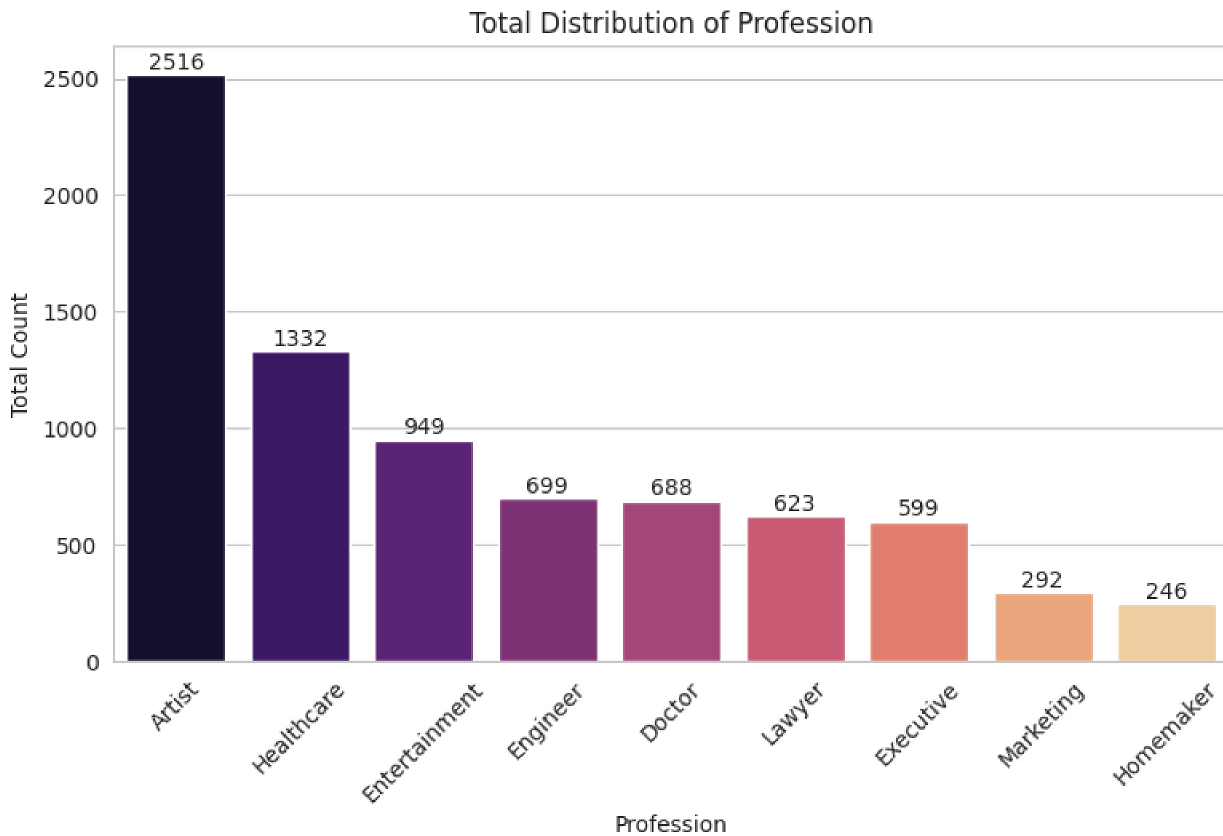


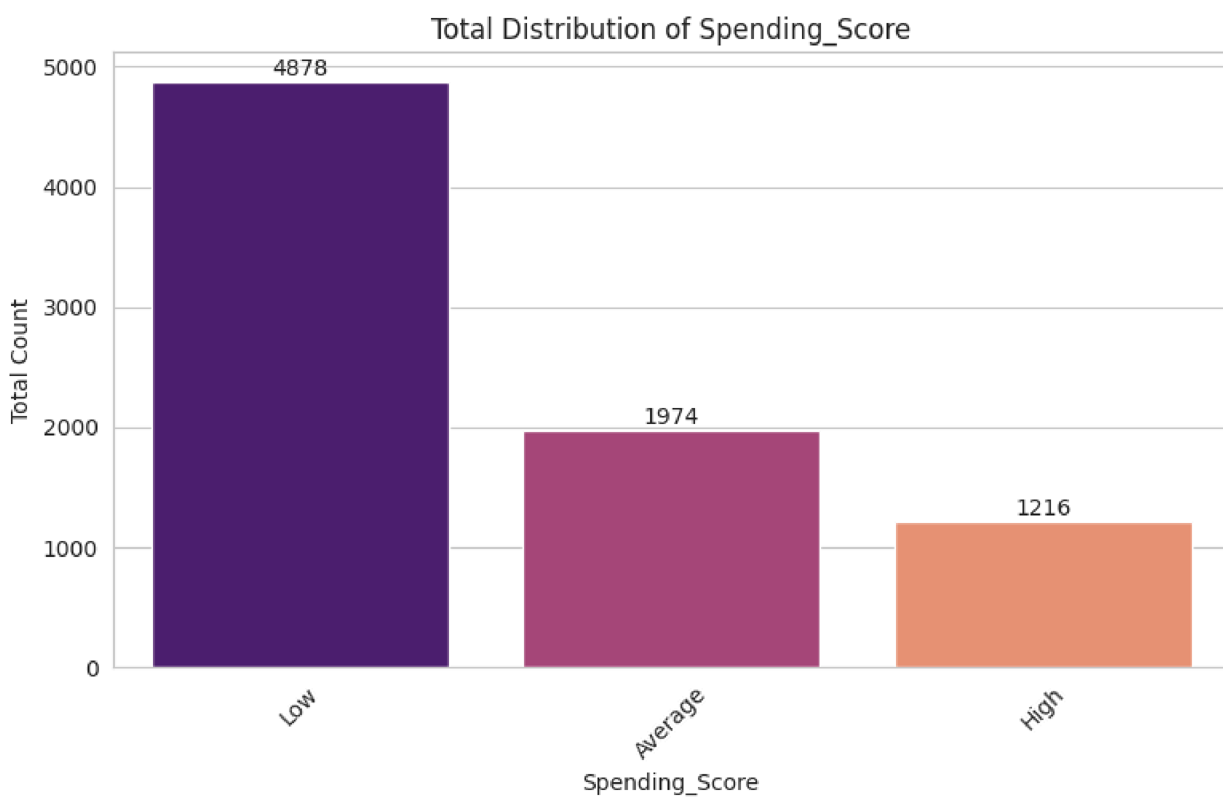
Barchart distribution of Categorical Features:

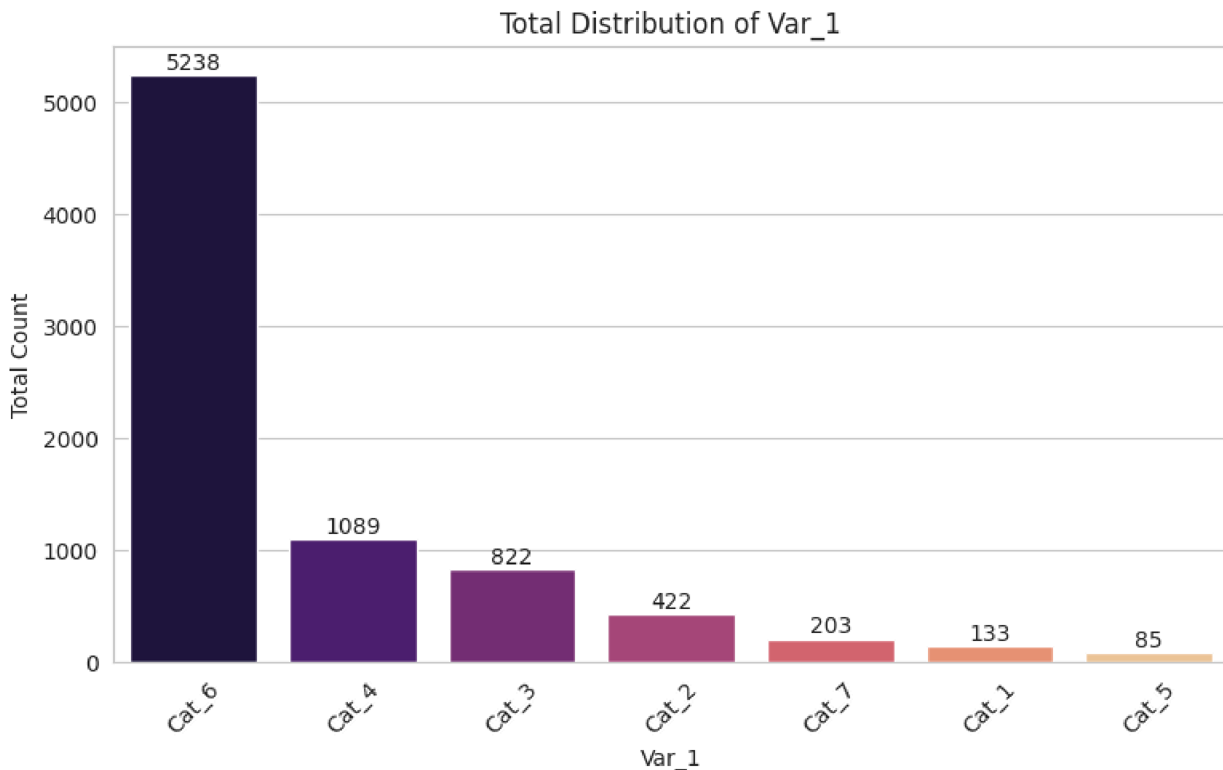



Total Distribution of Ever_Married
4643
3285
4000
3000
2000
1000
0
Total Count
Yes
No
Ever_Married

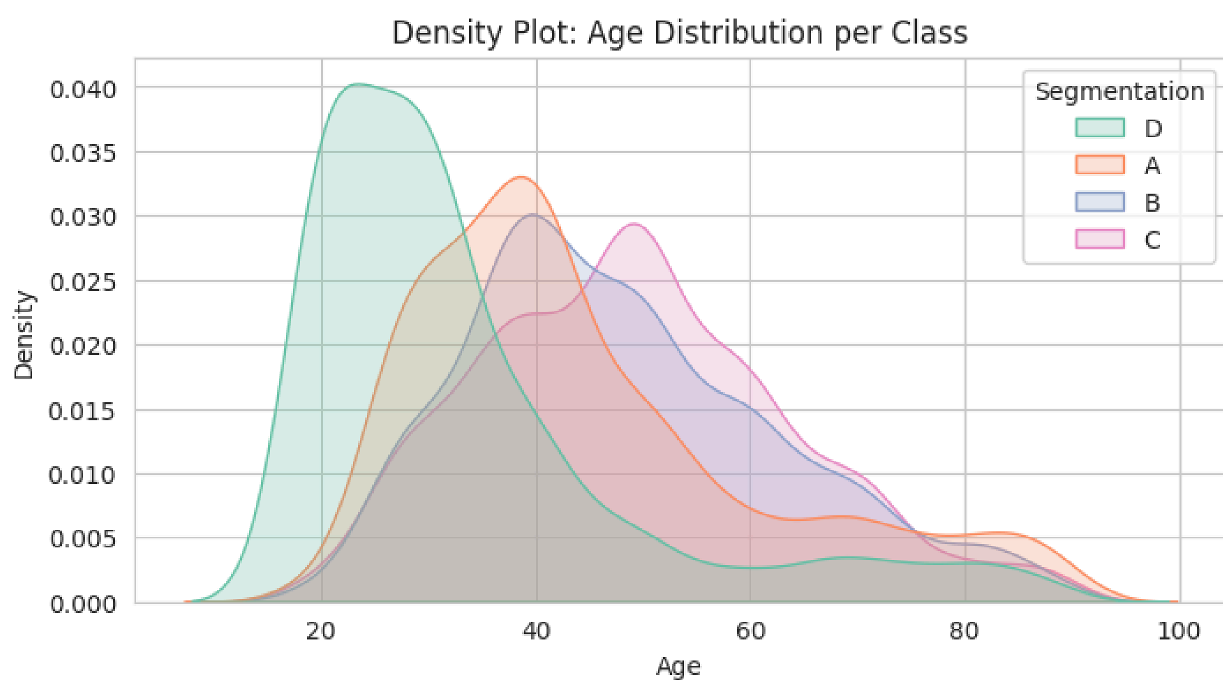

Total Distribution of Graduated
4968
3022
5000
4000
3000
2000
1000
0
Total Count
Yes
No
Graduated

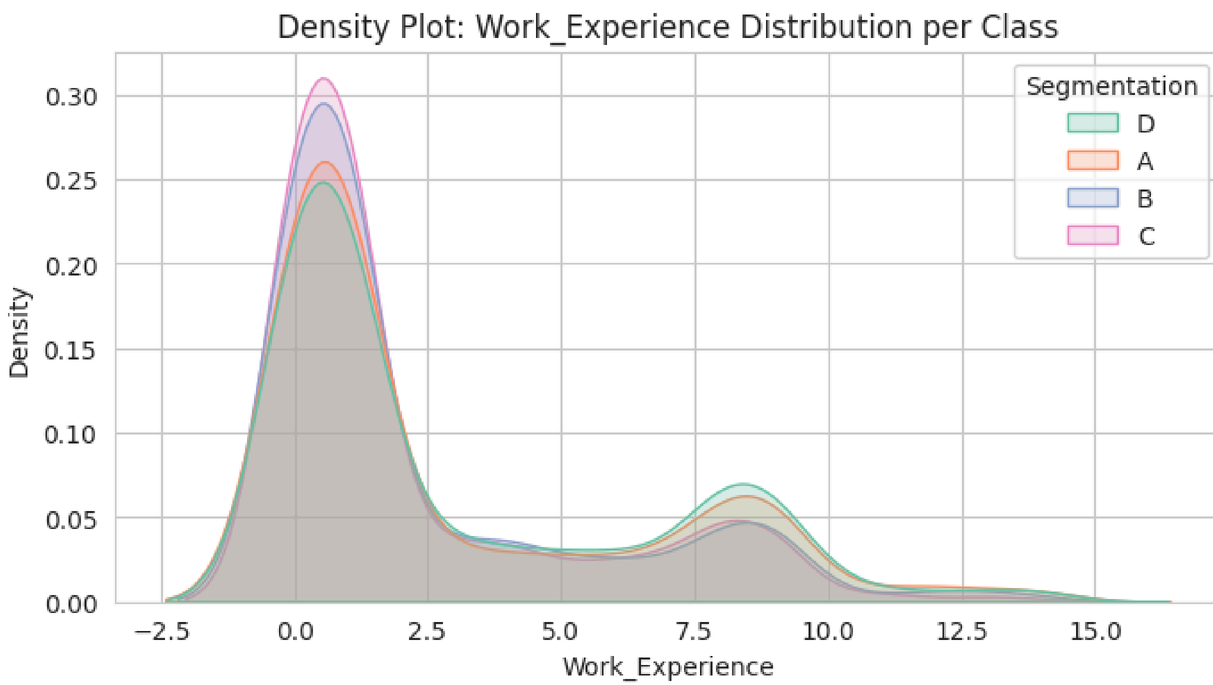

Total Distribution of Profession
Total Count
2500
2000
1500
1000
500
0
2516
1332
949
699
688
623
599
292
246
Artist
Healthcare
Entertainment
Engineer
Doctor
Lawyer
Executive
Marketing
Homemaker
Profession

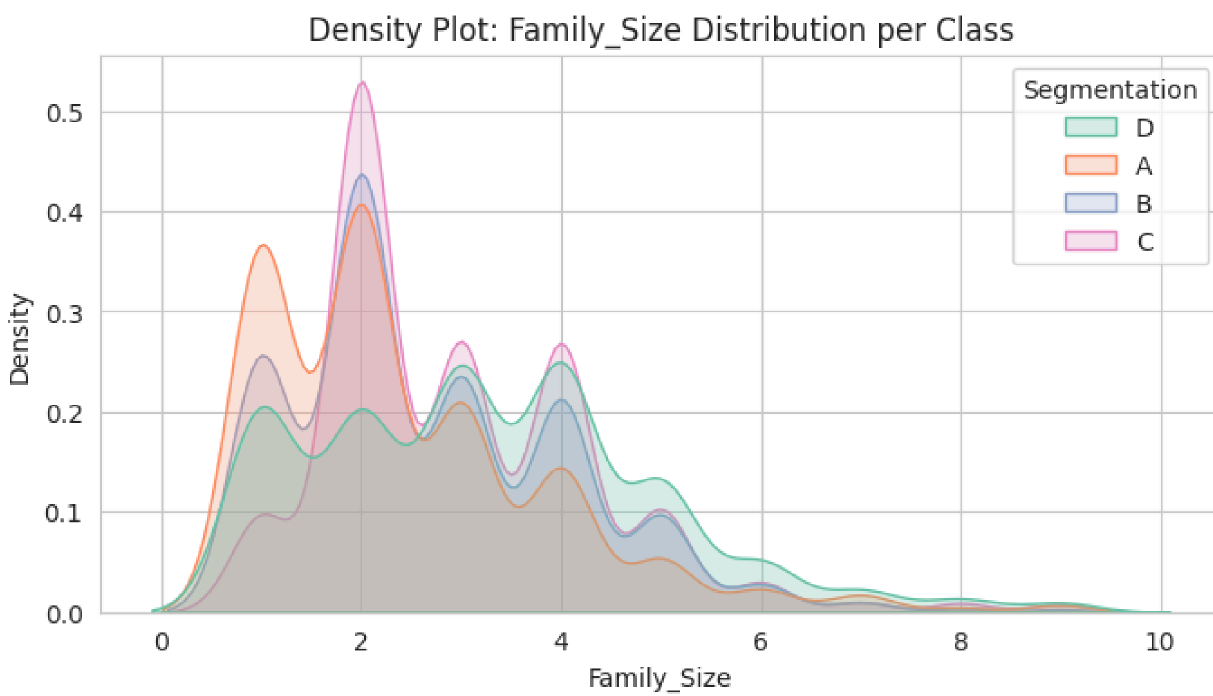

Total Distribution of Spending_Score
Total Count
5000
4000
3000
2000
1000
0
4878
1974
1216
Low
Average
High
Spending_Score



Density plot of Numerical data distribution per Segmentation:



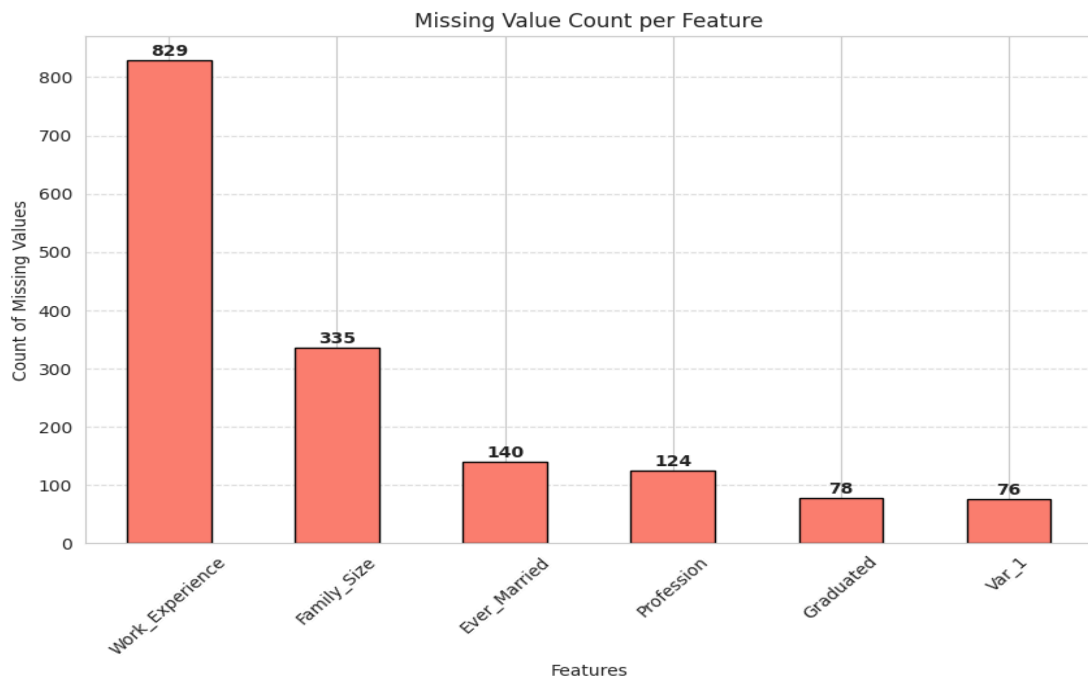

Density Plot: Work_Experience Distribution per Class
Segmentation
D
A
B
C
Density
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
Work_Experience


Density Plot: Family_Size Distribution per Class
Segmentation
D
A
B
C
Density
0.5
0.4
0.3
0.2
0.1
0.0
0
2
4
6
8
10
Family_Size

3. Dataset Pre-processing

Problem 1: Null / Missing Values

Several features contained missing values. The highest numbers of missing data were found in Work_Experience (829) and Family_Size (335), with smaller amounts in categorical features such as Ever_Married (140), Profession (124), Graduated (78), and Var_1 (76). SInce machine learning models can not handle null values, they had to be handled.



Solution: To prevent the loss of valuable data points that would occur from simply dropping rows, imputation techniques were applied.

Numerical features with missing values were imputed using the median value of their respective columns because they are right skewed.

Categorical missing values were imputed using the mode (the most frequent category).

Problem 2: Categorical Values

The dataset contains several input features in string format (Gender, Ever_Married, Graduated, Profession, Spending_Score, Var_1, Segmentation), which machine learning models cannot process directly.

Solution: Encoding was applied based on the nature of the categories:

- **Binary Label Encoding:** Applied to features with only two categories (Gender, Ever_Married, Graduated) to map them efficiently to 0s and 1s.
- **One-Hot Encoding:** Applied to multi-class features (Profession, Spending_Score, Var_1). This converted them into binary matrices, preventing the models from incorrectly assuming any mathematical or ordinal relationship between distinct categories.
- The target variable, Segmentation was encoded as well using Label encoder.

Problem 3: Categorical Values

The column **"ID"** was used only to represent unique identifiers for each customer record. Since a customer's ID does not contain meaningful information related to customer segmentation, it cannot contribute to predicting the target segment.

Solution:

The **ID** column was removed from the dataset to avoid introducing unnecessary or misleading information into the machine learning models.

4. Dataset Splitting & Feature Scaling

1. Dataset Splitting Before any scaling was applied, the pre-processed dataset was split to ensure the models could be evaluated on entirely unseen data.

- **Train set (80%):** Used to train the machine learning models.
- **Test set (20%):** Used for evaluating the performance metrics of models.

2. Feature Scaling

The numerical features (Age, Work_Experience, Family_Size) existed on vastly different scales. Leaving them unscaled would cause gradient descent algorithms and distance-based algorithms to give more importance to larger numbers heavily, leading to biased training.

So**,** StandardScaler was applied to these specific continuous features to give them a mean of 0 and a standard deviation of 1. The scaler was only fitted to the training data (fit_transform). The testing data was then scaled using those learned parameters (transform).

5. Model Training (Supervised and unsupervised)

Supervised Training:

The pre-processed dataset was trained using five supervised machine learning algorithms to identify the best fit for our multi-class customer segmentation task. The models deployed were:

- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest
- Logistic Regression
- Neural Network

Unsupervised Training:

As per the requirements of the project, the problem was approached as an unsupervised learning task. The first step involved determining the optimal number of clusters for the dataset. To achieve this, clustering experiments were conducted with values of *k* ranging from 1 to 10.

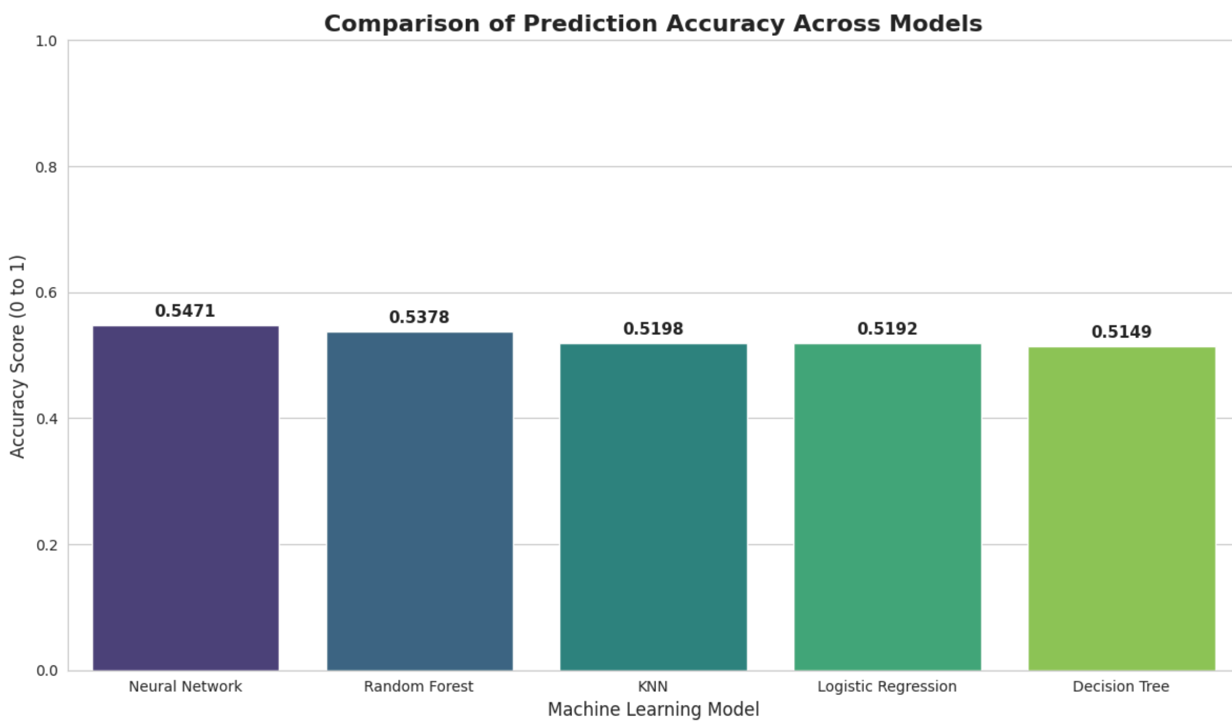
Using the Elbow Method, it was observed that k = 3 provided the most suitable clustering structure for the data.

After identifying the optimal number of clusters, the unsupervised clustering model was trained to classify customers into distinct groups based on their underlying patterns and similarities.

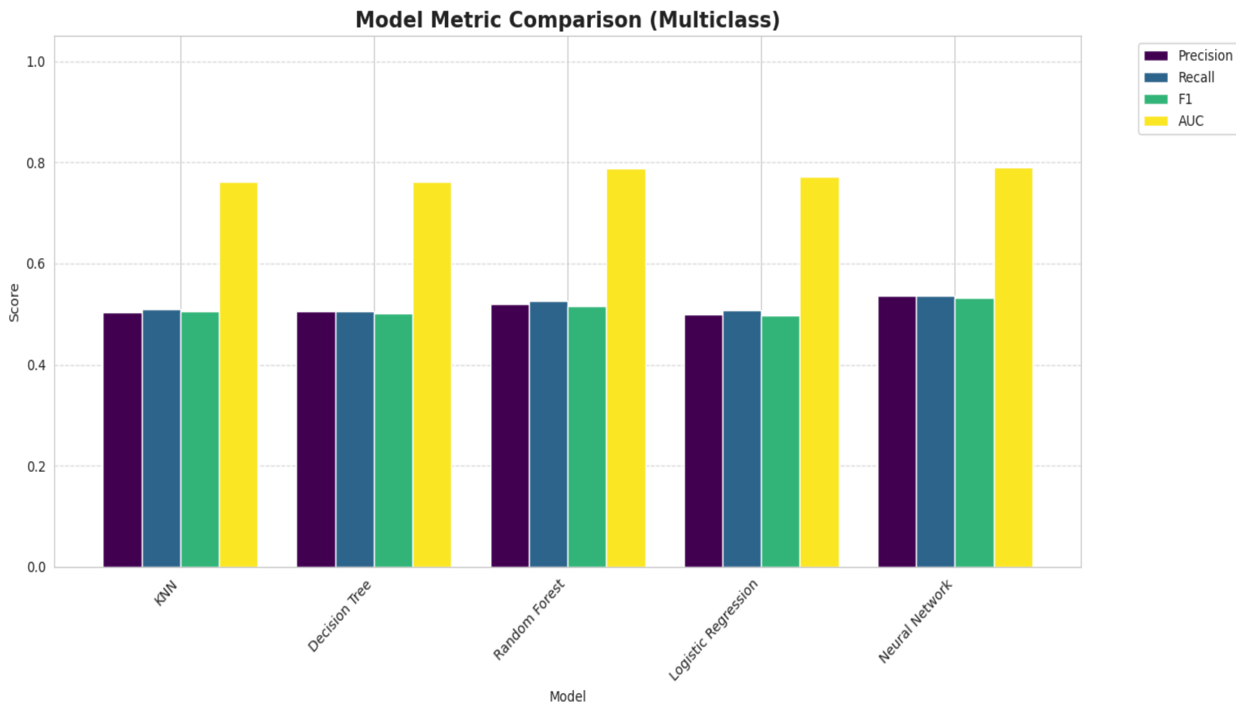
6. Model Comparison Analysis

The models were systematically evaluated and compared using their accuracy, precision, recall, F1-score, and AUC (Area Under the Curve) scores.

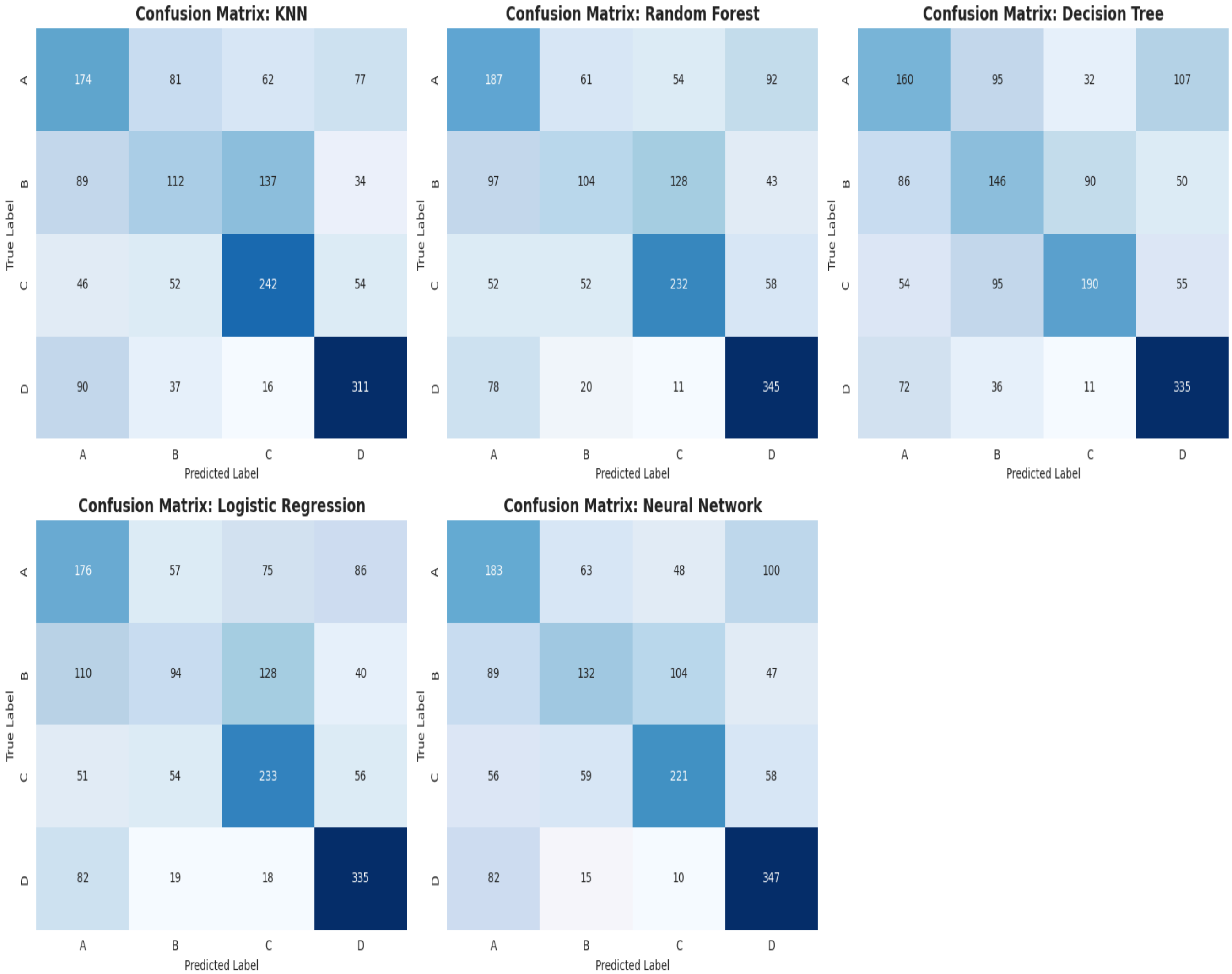
Bar chart showcasing prediction accuracy of all models:



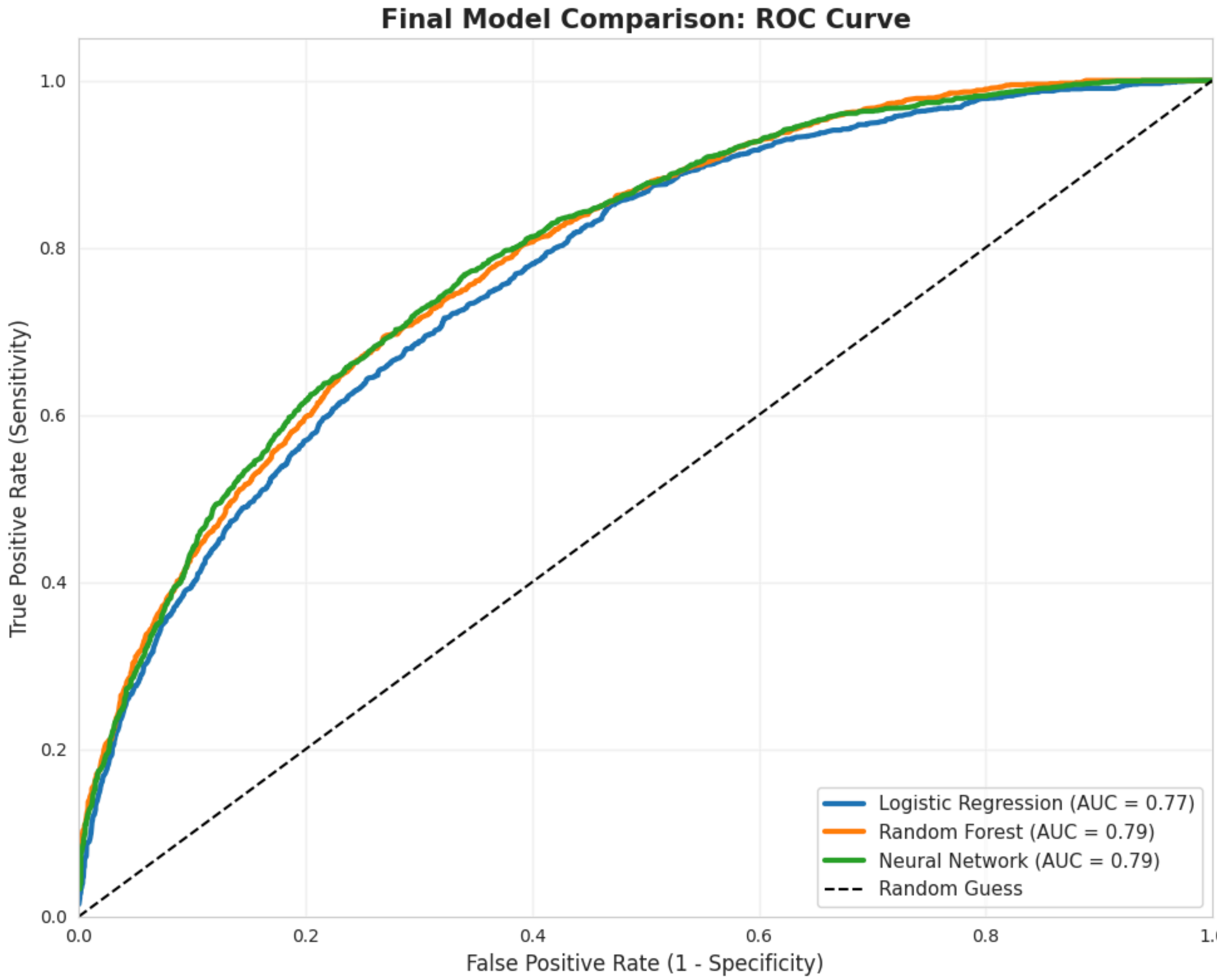
Precision, recall, F1 score comparison of each model:



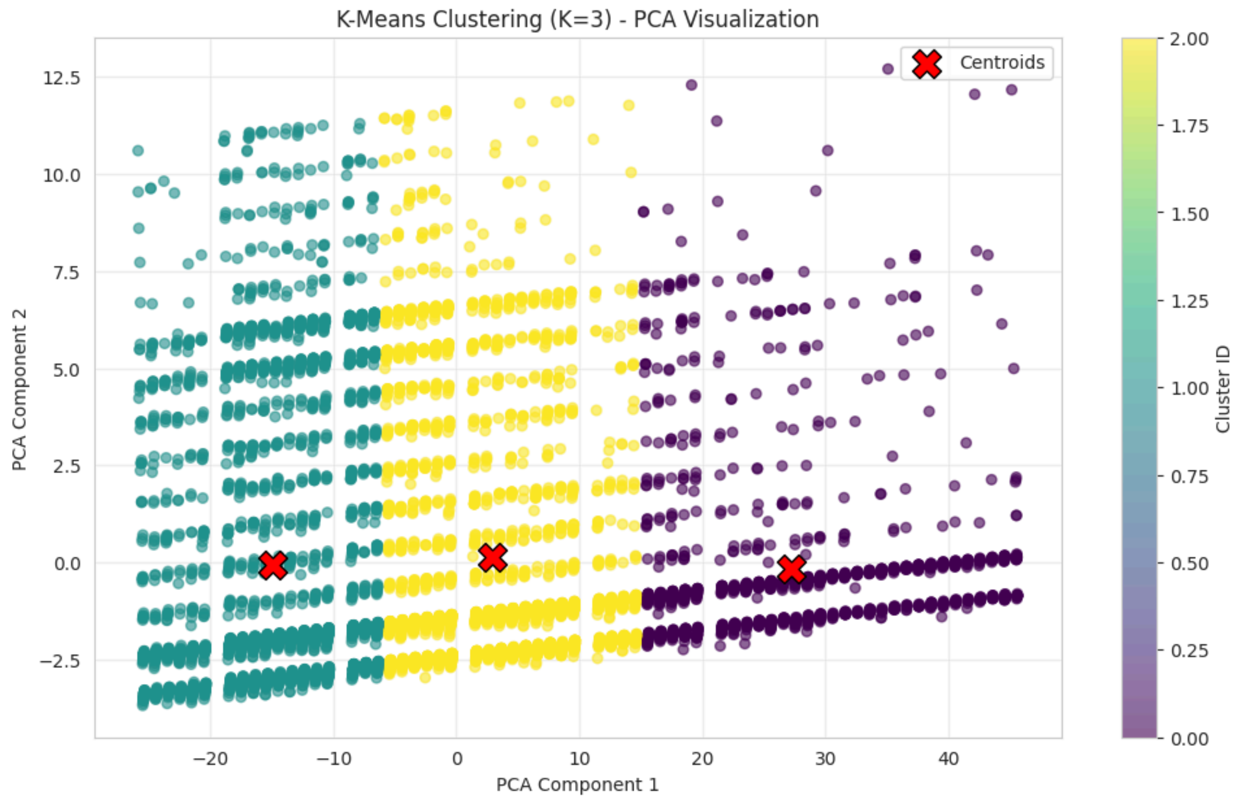
Confusion Matrices:



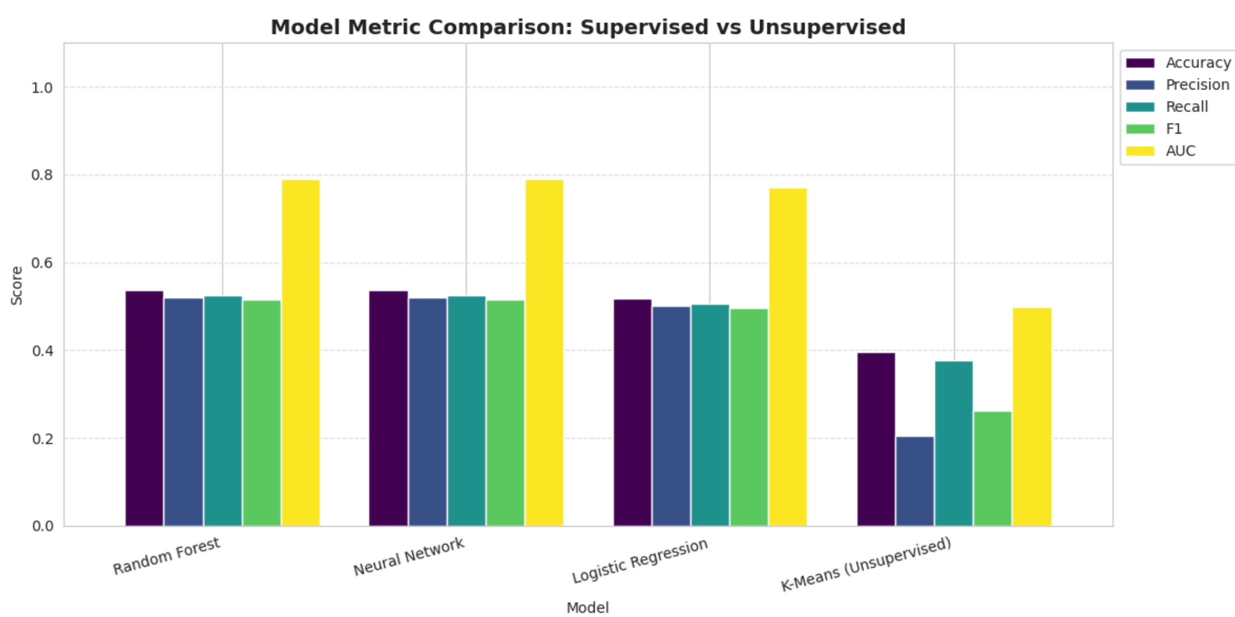
AUC score, ROC curve:



Unsupervised Model Result:



Supervised vs Unsupervised Result:



Final Score:

Model	Precision	Recall	F1 Score	AUC
Neural Network	0.535674	0.536134	0.531449	0.791095
Random Forest	0.519414	0.525733	0.516631	0.789358
Decision Tree	0.505812	0.504671	0.502386	0.761428
KNN	0.504039	0.510484	0.504727	0.761653
Logistic Regression	0.500219	0.507161	0.496985	0.771636
K-Means (Unsupervised)	0.205743	0.378319	0.263332	0.500000

6.1 Performance Analysis:

The performance evaluation of the machine learning models shows differences in their ability to classify customers into the target segments. Among all the models, the Neural Network achieved the best overall performance, obtaining the highest Precision (0.535674), Recall (0.536134), F1 Score (0.531449), and AUC (0.791095). These results indicate that the Neural Network was the most effective model in identifying customer segments while maintaining a balance between precision and recall.

The Random Forest model gave the second-best performance with an F1 Score of 0.516631 and an AUC of 0.789358, which was very close to the Neural Network. Its strong performance shows the effectiveness of capturing non-linear relationships within the dataset.

The Decision Tree and K-Nearest Neighbors (KNN) models showed moderate performance. Both models achieved similar results, with F1 scores around 0.50 and AUC values near 0.76. While these models were able to classify customer segments reasonably well, their predictive capability was lower compared to the Neural Network and Random Forest models.

The Logistic Regression model produced slightly lower performance than Decision Tree and KNN, with an F1 Score of 0.496985. Since Logistic Regression is a linear model, its relatively weaker performance suggests that the dataset contains complex and non-linear relationships (which confirms heatmap interpretation) that cannot be fully captured using a simple linear decision boundary.

The K-Means clustering model, treated as an unsupervised learning approach, showed the weakest performance overall. It achieved a very low Recall (0.378319) and F1 Score (0.263332). This result was pretty much expected because the best k was found by the algorithm was 3 while there were actually 4 classes. Consequently, supervised learning models significantly outperformed the unsupervised approach in this project.

7. Conclusion

Understanding from the Result:

The results of this project show that machine learning models can effectively classify customers into different segments (A, B, C, and D). Among all the models, the Neural Network achieved the best overall performance, followed by the Random Forest model. This indicates that advanced supervised learning techniques are more capable of identifying complex patterns between customer features and their corresponding segments.

The moderate performance of models such as Decision Tree, KNN, and Logistic Regression suggests that the customer segments share some characteristics which are overlapping, making the classification task more challenging. On the other hand, the K-Means clustering model performed poorly because the naturally formed clusters did not closely match the predefined customer segments. It could only identify 3 clusters while in reality, there were 4 classes.

Reason behind getting such results:

One reason for these results is that customer segmentation data contains non-linear relationships and complex feature patterns that simpler models cannot fully capture. In contrast, Neural Networks and Random Forests are better at learning these hidden and non linear relationships, which helped them achieve higher accuracy and F1 scores.

Challenges Faced:

One of the challenges during the project was designing the Neural Network architecture. It was difficult to determine the optimal number of hidden layers, neurons, and activation functions. Different combinations were tested manually to identify the best-performing configuration while also ensuring that the model did not overfit due to having many layers.

Another challenge occurred during implementation when parts of the Random Forest and Neural Network codes were accidentally mixed, causing both models to produce identical evaluation metrics. This created confusion during the analysis phase and required debugging to identify and fix the issue. Additionally, interpreting the best k from the graph using the elbow method was difficult.